- ENGINEERING MATERIALS -

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MODULE II : Ceramics

Sr. No.	TOPIC	REMARKS
1.	A) Introduction of Ceramics – Definition, types, properties and uses.	Types of ceramics materials, Classification on the basics of sources a) Natural occurring b) Ceramics produced by Heat treatment methods. properties and uses.
2.	B) Glass – Definition, Properties, Types with uses.	brief information and uses of each type of glass. Types of glass: Soda lime glass, Borosilicate glass, Lead Glass, Alumina-Silicate Glass, Laminated Glass, Optical Glass.
3.	C) Abrasives – Natural and Artificial Abrasives – Examples, Properties and Uses	Introduction to abrasives Examples can be mentioned with the types; Properties, uses

Sr. No.	TOPIC	REMARKS
4.	D) Optical fibres – Definition, Components of optical transmission system, Advantages of optical fibre communication, Applications of glass-based fibre - optical fibres	Introduction Optical Fibers derived from Ceramics only to be taught Components; Core, Cladding, Jacket with their function. Advantages and applications of glass-based fibre optical fibre.

Reference Books (Ceramics):

1. Optical Fiber Communications (fourth edition)- Gerd Keiser—Mc Graw Hill publication

2. A Textbook of Engineering Chemistry- Dr. Sunita Rattan- S.K. Kataria and Sons.

MODULE II : Ceramics

- Ceramics are defined as inorganic, crystalline, non-metallic materials that are processed and/or used at high temperatures.
- Most ceramic materials contain silicates, metallic oxides and their combinations.
- E.g. clay, carborundum, boron nitride, cement, brick, etc.

Types of Ceramic materials

- Natural occurring: These materials are used to make traditional ceramics, which are used in household items and building materials.

Examples: clay, silica, feldspar, and other silicates

Clay: It is made up of hydrated aluminum silicates. It is used to make structural clay products like bricks, tiles, and pipes.

Silica: It is derived from natural silica minerals. These ceramics are often used in glass makings and refractories.

Feldspar: A common ceramic raw material that acts as a fluxing agent to reduce the melting temperature of aluminosilicates.

Types of Ceramic materials

- Ceramics produced by Heat treatment methods:

1. Glass-ceramics: Produced by controlled crystallization of glass through heat treatment. Used in making cookware, dental implants, aerospace, medical devices, electronics.
2. Sintered ceramics: Formed by compacting ceramic powders and then heating them to high temperatures to bond the particles. Used in making cutting tools, abrasives, catalytic converters.
3. Fired ceramics: Created by shaping clay or other ceramic materials and then firing them in a kiln at high temperatures. Used in making tablewares, sanitarywares, bricks and tiles.

Types of Ceramic materials

- Ceramics produced by Heat treatment methods:

4. Pyro ceramics: Produced by heat-treating ceramic materials at extremely high temperatures, often above 1000°C. They are used in making ceramic armor, high temperature coatings.

5. Ceramic matrix composites: Formed by infiltrating a ceramic matrix with a metal or polymer, and then heat-treating the composite. Used in making automative parts, energy storage, industrial processes.

GLASS

- Glass is an amorphous, hard, brittle, transparent or translucent, super-cooled liquid of infinite viscosity, obtained by fusing a mixture of a number of metallic silicates, most commonly of Na, K, Ca, and Pb.

Properties:

- Amorphous
- Has no definite melting point
- Can absorb, reflect or transmit light
- Can take high polish
- Is a good electrical insulator
- Is affected by alkalis
- Very brittle

TYPES OF GLASSES

Soda lime glass:

- The raw materials are silica (sand), calcium carbonate, and soda ash. Their approximate composition is $Na_2O.CaO.6SiO_2$.
- They are low in cost, resistant to devitrification, and relatively resistant to water.
- They melt easily and hence, can be hot-worked easily.
- These acids can be attacked by acids.
- Uses: they are widely used as window glasses, electric bulbs, plate-glasses, bottles, jars, building blocks, etc.

TYPES OF GLASSES

Borosilicate glass:

- It is the most common of the hard glasses of commerce.
- It contain silica and boron, with a small amount of alumina and some alkali oxides.
- A typical formula for such a glass would be following:

%	80.5	13	3	3	0.5
Component	SiO_2	B_2O_3	Al_2O_3	K_2O	Na_2O

- This glass has low thermal coefficient of expansion, and high chemical resistance.
- They have very high softening points and excellent resistivity.
- Uses: they are used extensively in industry for pipelines for corrosive liquids, gauge glasses, superior lab apparatus. Kitchenware, chemical plants, etc.

TYPES OF GLASSES

Lead Glass:

- It is made by using lead oxide, instead of calcium oxide, for fusing together with silica.
- Its approximate composition is $K_2O.PbO.6SiO_2$.
- Lead glass is more expensive than soda-glass and also a higher refractive-index.
- It has excellent electrical properties.
- It is bright, lustrous, and possesses high specific gravity (3-3.3).
- Uses: used for high-quality tableware, optical purposes (like lenses), neon sign tubings, cathode-ray tubes, electrical insulators and in art objects, because of their high lustre.

TYPES OF GLASSES

Aluminosilicate glass:

- A typical formula for such a glass would be following:

%	55	7	23	1	9	5
Component	SiO_2	B_2O_3	Al_2O_3	$Na_2O + K_2O$	MgO	CaO

- This glass possess exceptionally high softening temperatures.
- Aluminosilicate glass is strong and can resist scratches.
- It is highly resistant to chemical corrosion.
- Uses: for high pressure mercury discharge tubes, chemical combustion tubes, certain domestic equipments, etc.

TYPES OF GLASSES

Laminated glass:

- It is made by pressing (or bonding) together two or more sheets/plates of glass with one or more alternating layer of bonding material like plastic resin, asphalt or synthetic rubber.
- The laminated glass is: (i) shatter-proof (i.e., its pieces do not fly off, when suddenly broken); (ii) shock-proof (i.e., it can withstand sudden changes of temperature and even some pressure, without breaking).
- A bullet resistant laminated glass is obtained by pressing together several layers of glass with vinyl resins in alternate layers. Ordinarily, thickness of such glass varies from 12.7-76.5mm.
- Uses: as an ideal material for use as safety glass in aircrafts, helicopters, submarines, etc. Bullet-resistant laminated glass finds application in making automobile wind screens, looking windows, etc.

TYPES OF GLASSES

Optical glasses:

- It contain phosphrous and lead silicate, together with a little cerium oxide. The latter is capable of absorbing uv light. Moreover, they are given thorough homogeneity.
- This is secured by a very careful manufacturing process and by careful heating the molten glass for a prolonged time.
- In general, optical glasses have low melting-points and are relatively soft.
- Their chemical-resistance and durability are appreciably lower than those of ordinary glass.
- Uses: They are used making lenses.

Abrasives

- Abrasives are substances characterized by their hardness, and which are used to wear down softer surfaces by cutting, grinding or polishing.
- For example, the housewife uses abrasive stone to keep her kitchen knives sharp, the dentist uses an abrasive powder when he cleans teeth.

Types of Abrasives

- Natural Abrasives –

Diamond: it is crystallized carbon, ranks first in hardness, since it can cut into the surface of any other substance; whereas only another diamond can scratch a diamond.

1. It is chemically inactive and not attacked by acids, alkalis, or fused potassium chlorate.
2. Diamonds that are off-colour or otherwise faulty, are called borts.
3. The black diamond from Brazil, is called carbonado and has no value for jewellery.
4. Borts and carbonado are used in drill-points, as saw-teeth for cutting rocks, stones or grinding wheels, dressers, and in operations, where great hardness is required.

Types of Abrasives

- Natural Abrasives –

Corundum:

- It is crystallized aluminium oxide, comes next to diamond in hardness.
- It is not of gem quality, finds use as abrasives for grinding glasses, gems, lenses, metals and metal cutting.

Types of Abrasives

- Natural Abrasives –

Emery:

- It is fine-grained, opaque, massive mineral, dark-grey to black in color. It is an aggregate of 55-75% crystalline alumina, 20-40% magnetite (Fe_3O_4).
- Its hardness is about 8 on Mohs scale.
- The grinding power of emery depends almost entirely on the proportion of alumina it contains, and on the effect of the other ingredients in determining true hardness of this mineral.
- Emery is used as tips of bits of cutting, and drilling tools, etc. Artificial emery used for buffing and polishing is a graded combination of alumina and magnetite.

Types of Abrasives

- Natural Abrasives –

Garnets:

- These are trisilicates of alumina, magnesia, and ferrous oxide. The common garnet used as an abrasive is a complex calium-aluminium-iron silicates.
- Hardness of garnets ranges from 6-7.5 on Mohs scale.
- Garnets are too soft for grinding steel and iron, but when glued to paper or cloth, they are used for finishing hard woods.
- They are also used for bearing pivots in watches, glass-grinding, and polishing metals.

Types of Abrasives

- Artificial Abrasives –

Carborundum or SiC:

- It is a bluish-black crystalline artificial mineral with hardness between carborundum and diamond.
- It is made by subjecting a mixture of silica and carbon to high temperature (1650-2200 ^{0}C) in an electric furnace.

$$SiO_2 + 3C \rightarrow SiC + 2CO \uparrow$$

- It is very hard (Mohs scale 9.3), chemically inactive, can withstand the action of temperature without damage.
- It is mainly used in cutting wheels, abrasive papers, and clothes.

Types of Abrasives

- Artificial Abrasives –

Alundum (Al_2O_3):

- It is prepared by subjecting a mixture of subjecting a mixture of calcinated bauxite, coke, and iron to high temperature. In electric arc furnace.
- It is less brittle, and tougher.
- It is used for grinding steels and other material at high tensile-strengths.

Types of Abrasives

- Artificial Abrasives –

Boron carbide (B_4C):

- It is inert and one of the hardest (about 9 Mohs) of all artificial abrasives, and made by heating boron oxide with coke in an electric furnace to approximately 2700 ^{0}C.
- $$B_2O_3 + 7C \rightarrow B_4C + 6CO \uparrow$$
- It is used on hard materials for making grinding dies, and for cutting and sharpening hard high-speed tools.

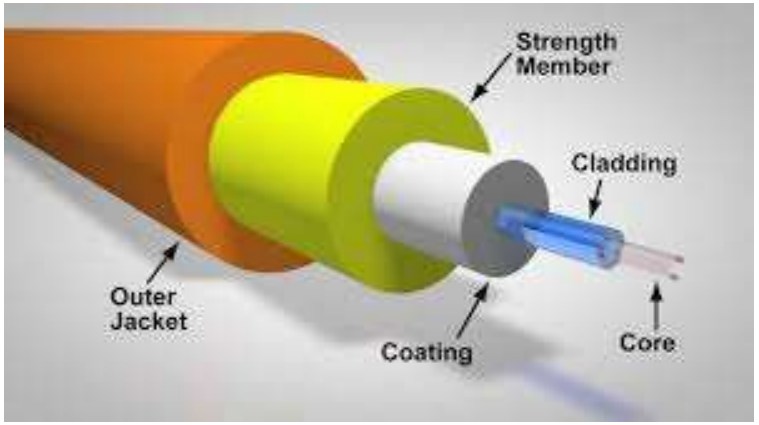
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Optical Fibres

- Optical fibers are thin, flexible strands of glass or plastic that transmit data as light signals.

- Components of Optical Fibres:

1. Core,
2. Cladding, and
3. Jacket (coating).



Components of Optical Fibres

- **CORE**:

Definition: The central part of the optical fiber where the light signal is transmitted usually made of glass or plastic.

Function: The core is made of a thin glass or plastic center that carries the light signal (provides a pathway for light to travel). It has a higher refractive index than the cladding.

The core is responsible for transmitting the light signal through the fiber.

Components of Optical Fibres

- **Cladding**:

Definition: The outer layer that surrounds the core.

Function: The cladding is made of a material with a lower refractive index than the core. It helps to keep the light signal within the core by reflecting it back.

The cladding is responsible for containing the light signal within the core and preventing it from escaping.

Components of Optical Fibres

- **Jacket**:

Definition: The outermost layer of the optical fiber.

Function: The jacket is a protective layer that covers the fiber and provides mechanical strength.

The jacket protects the fiber from environmental factors such as moisture, temperature, and physical damage.

Advantages of glass-based fibre optical fibers

- High Bandwidth: Glass-based fibers offer high bandwidth, enabling fast data transmission rates.
- Low Attenuation: Signals transmitted through glass fibers experience low attenuation, ensuring reliable data transmission over long distances.
- Immunity to Interference: Glass fibers are resistant to electromagnetic interference, reducing errors and data corruption.
- Security: Glass fibers are difficult to tap, providing a secure transmission medium.

Advantages of glass-based fibre optical fibers

- High-Speed Data Transmission: Glass fibers support high-speed data transmission, making them ideal for applications requiring fast data transfer.
- Long-Distance Transmission: Glass fibers enable long-distance transmission without signal degradation, reducing the need for repeaters.
- Resistance to Environmental Factors: Glass fibers are resistant to environmental factors like temperature, humidity, and vibration.
- Low Power Consumption: Glass fibers require low power consumption, reducing energy costs and environmental impact.

Applications of glass-based optical fibre

- Telecommunications: Long-distance phone and internet connectivity.
- Internet Backbone: High-speed data transmission across continents.
- Cable Television: Digital TV transmission.
- Medical Applications: Endoscopy, surgical procedures, and medical imaging.
- Sensing Applications: Temperature, pressure, and strain sensing.
- Aerospace and Defense: Secure communication, navigation, and sensing.
- Data Centers: High-speed data transmission and storage.


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